Human cognition project

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Introduction

This project aims to investigate how human memory recalls information by conducting and analyzing a series of behavioural experiments. According to previous research, in free recall tasks, people typically remember the first and last items in a list better than those in the middle[1, p8, L185-205], known as the primacy and recency effects, reflecting long-term and short-term encoding, respectively. In serial recall tasks, where the order does matter, performance should reveal the limited capacity of working memory should usually be around 6-7 items. By comparing recall accuracy across conditions, we examine how presentation speed and disruptions to working memory influence performance.

Method

The experiment was conducted on a group of 4 students (us) who each completed a set of memory tasks. Each of these tasks were performed 20 repetitions per condition, resulting in a total of 160 trials per participant.

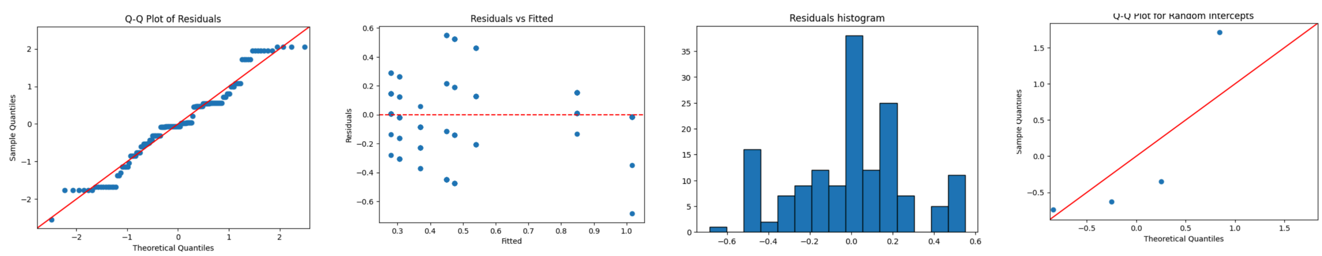
The tasks consisted of sequences of ten common words (e.g. “cat”, “star” and “pen”) or single letters presented sequentially. Whereas the words were randomly drawn from a predefined list to minimize predictability, this list can be viewed within the code. A custom Python program automated the testing workflow. This allowed configuration of session parameters such as presentation mode (sequential or all at once), recall type (ordered or unordered), presentation rate, number of items, inclusion of math questions, a 30 second delay before recall, the use of letters instead of words. After each trial, the program automatically stored both the configuration settings and the participant’s responses in a CSV file and then prompted whether to restart with the same setup. An overview of the procedure can be viewed on the following table:

	Test	Repetition amount	Delay	Item amount	Other
Free recall (unordered)	1	20	2 s	10 words	
	2	20	0.75 s	10 words	
	3	20	2 s	10 words	Math question before answering
	4	20	2 s	10 words	30 s delay before answering
Serial recall (ordered)	5	20	2 s	10 words	
	6	20	2 s	10 words	Repeating tah-dah during the test
	7	20	2 s	10 words	Tapping fingers to ‘we will, we will rock you’ melody during the test
	8	20	2 s	10 letters	

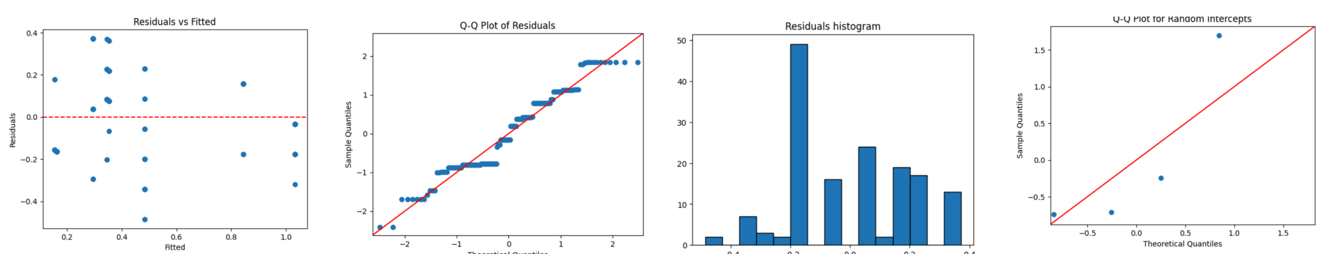
In regards to the data analysis, free recall performance was scored as the proportion of correctly recalled items overall and within four position groups: primacy (first 3 words), nonprimacy (last 7 words), recency (last 3), and non recency (first 7 words). For serial recall, accuracy was defined as the proportion of items recalled in the correct order, stored in percent. Mean proportions, confidence intervals, and t-tests were used to compare performance across conditions.

Results and discussion

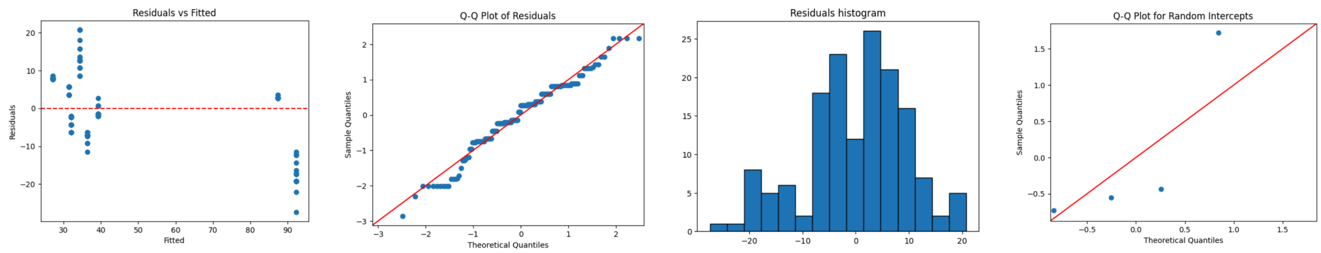
Primacy effect: To test the primacy effect, a mixed-effects linear model with a random intercept for participant was fitted: proportion correct ~ 1 + score_type(1|participant)[1].
Across 154 observations from four participants, the non-primacy mean was 0.451, and the primacy effect was 0.169 (SE: 0.044, z: 3.833, p<.001, 95%CI: [0.082;0.255]), which corresponds to the predicted primacy at around 62%. Furthermore, diagnostics showed that residuals were centered around zero with no systematic pattern, and Q–Q plots for both residuals and random intercepts followed the diagonal reasonably well, this is necessary to prove validity of the model use because the mixed-effects model assumes: normality of residuals, homoscedasticity, random distribution of the residuals, normal distribution of random effects (altough hard to prove with this few independent clusters).
This significant difference at 16.9% shows that words that are presented at the start of a list recall better than the middle ones. The effect is quite substantial with an improvement of around 37% compared to the baseline of 45.1%. Therefore this correctly aligns with the previous research, that early sequence means that the working memory is less loaded which therefore allows more cognitive resources to be devoted to encoding items into long term memory. [1,p8, L190-205]



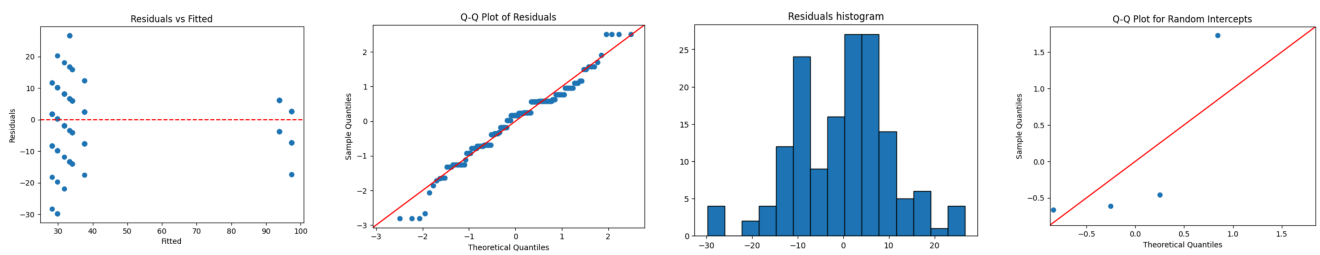
Latency effect: To test the recency effect in a free recall task, the same procedure as for the primacy effect was used, but with focus on the last 3 items instead of the first 3.
Across the same 154 observations the non-recency mean was 55.4%, and the recency effect was -0.19 (SE: 0.03, z: -5.76, p<.001, 95%CI: [-0.26;-0.13]), corresponding to the predicted recency of 36.4%, Thus the last items were recalled 19% worse than the rest.
The negative recency effect is statistically significant (p < 0.001), and is therefore found to be the opposite of the research paper. Participants may have rehearsed early words or begun recall from the start of the list, reducing memory for the final items.[1, p8, L190-205]



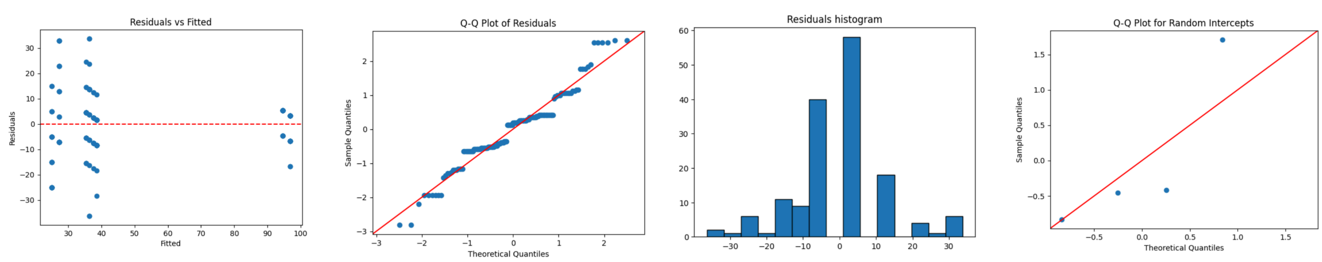
Presentation speed: Rather than the speed be 2 seconds, it gets shortened to just 0.75 seconds. When this happened, each participant remembered around 45% rather than the 50%. Although this difference may seem small (about 5 percentage points), it was statistically significant (SE = 1.56, z = 3.14, p = .002, 95% CI [1.85; 7.98]). This means that people remember more when they have extra time to look at each word. The slower speed lets them rehearse the first few words longer and store them in long-term memory.[1, p8, L190-205]



Math task or waiting before recall: Other two versions of the free recall include the addition of math tasks and waiting 30 seconds before the recall. The math question reduced the recall from 50% to around 46%, which corresponds to missing one more word out of ten. The change is small but statistically significant (SE = 1.73, z = −2.06, p =0 .039, 95% CI [−6.94; −0.17]). This result shows how even a very short task can disturb memory. The math problem kept the brain busy by blocking the working memory that would normally hold the last few words. This is why people often forget recent information, when they are distracted right after learning something. [1, p8, L205-210]

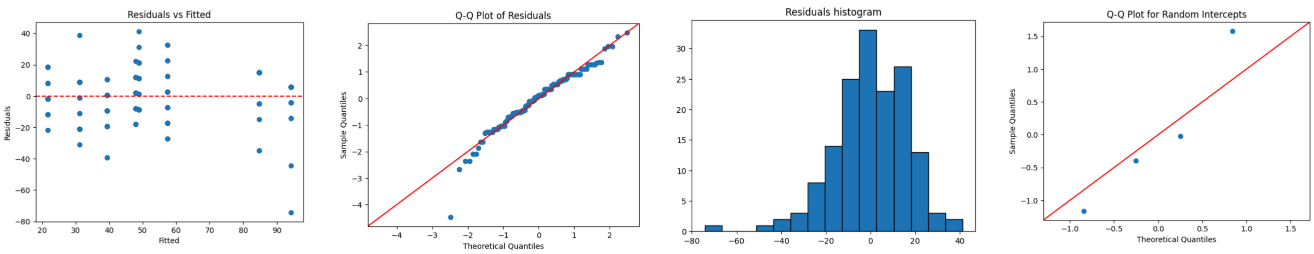


When introducing the 30 second delay without any tasks before recall, the recall stayed almost the same at 48% compared to the 50%. This means that the small drop was not statistically significant (SE = 2.09, z = −1.04, p = .298, 95% CI [−6.27; 1.92]). This shows that just waiting does not make people forget, as long as they do not use their working memory for something else during the delay. Forgetting happens mainly because of interference, and not from the passive fading of memory over time. [1,p8, L205-210]

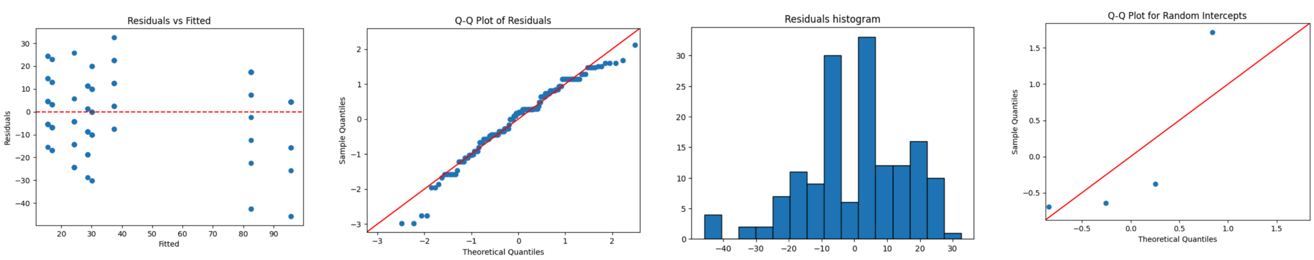


Serial recall and chunking: Now and further on, this experiment will be focusing on serial recalling, where the participants had to remember each of the items in order. This as expected was a much harder task to do, which also showed in the test result as it shows that on average the participants remembered 4 out of 10 words (41.9%, SE = 16.1, 95% CI [10.4%; 73.5%]). The mixed-effects model also showed substantial variation between participants (SD ≈ 32 percentage points), indicating that some individuals performed much better than others. Although the mean recall was below the well-known “seven-item” rule, the uncertainty in the data suggests that true capacity may still fall near that range. This supports the classic finding that working memory typically holds about seven ± two items. [2, p9, L20-45]

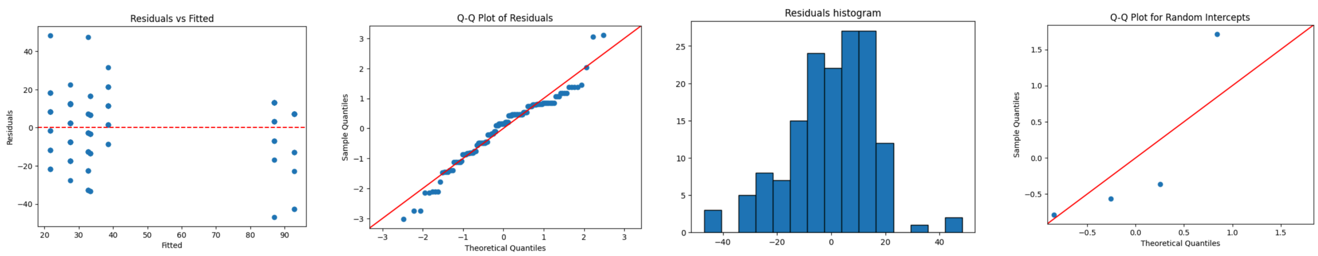
Although the serial recall task shows the limits of working memory, people can sometimes overcome these limits by grouping information into meaningful units, known as chunking. To test this, we compared lists of single letters with lists of three-letter words. Contrary to expectations, participants recalled 57.9% of the letters but only 48.4% of the words, a difference of −9.44 percentage points (SE = 2.71, z = −3.49, p < .001, 95% CI [−14.75; −4.14]). This outcome demonstrates the word length effect described in the lecture notes, longer words take more time to rehearse internally, leaving less capacity for other items. Because our words were random and unrelated, participants could not group them into meaningful chunks. True chunking only helps when there is a clear pattern or semantic connection, such as remembering “cat, dog, pig” as a group of animals. [2,p9, L70-95]



Speaking or finger tapping: Other two versions of the serial recall included speaking (repeating ta-dah out loud) and fingertapping (to the melody of: we will rock you), while recalling. In the case of speaking, the recall dropped sharply from 48.4% to 34.8%, which is a loss of more than 13% (SE = 2.51, z = 5.27, p < 0.001, 95% CI [8.30; 18.12]). This is likely because saying tah-dah blocks the use of the phonological loop, which is a part of the working memory that normally helps people repeat the words in the head. When this loop is blocked, people can no longer rehearse the list and their memory system shrinks from around 7 to only four items. [2, p9, L100-130]



Finally, we tested a control condition where participants tapped their fingers to “We Will Rock You” while trying to remember the words. Their recall accuracy dropped slightly from about 48% to 42%, a decrease of 5.9 percentage points (SE = 2.55, z = −2.32, p = 0.021, 95% CI [−10.88; −0.90]). Although statistically significant, this decline was much smaller than the effect of speaking, representing roughly a 12% relative reduction. This suggests that finger tapping caused mild distraction rather than genuine interference with the verbal memory system. As explained in the lecture notes, nonverbal tasks like tapping usually have little impact on verbal working memory because they rely on different neural processes. [2, p10, L135-160]



Recurring: When looking at the errors gives more clues about how memory works. Many participants swapped the order of nearby words, a common transposition error, showing that people remember the items but not always their exact order. Some also mixed up letters that sounded alike, such as “B” and “P.” These phonological errors show that working memory keeps information in a sound-based form even when the items are shown visually. [2, p9, L100-120] Small spelling mistakes also appeared, but were random and unimportant.

Altogether, these findings support the dual-system model of memory described in the lecture notes. Working memory is short-term, limited and easily disrupted, while long-term memory builds slowly, but lasts longer. The primacy effect shows information being transferred into long-term memory, whereas the recency effect (when it appears) reflects items still active in the phonological loop. Anything that limits rehearsal, like faster presentation, a second task or speaking aloud, reduces recall, while simple waiting or a nonverbal task has little effect. Even with a small sample, the effects were strong and clear, showing how simple experiments can reveal the structure of human memory and the processes that decide what we remember and what we forget.

Conclusion

In our eight experiments, we were able to repeat several well-known findings about how human memory works. The participants clearly showed a primacy effect, meaning they remembered the first words in a list best. They also remembered more when the words were shown slowly and performed much worse when they had to speak (“tah-dah”) at the same time, which blocked their ability to rehearse the words in their head. The short math task reduced memory for the last few items, while a simple waiting period did not, showing that distraction, rather than time itself, is what causes forgetting. In the serial recall task, people could remember about four out of ten words, which fits within the expected range for working memory. Finally, chunking only helped when the information had clear meaning or structure. It is however important to note that these conclusions are drawn from data from only four people, which is not a large enough sample size to accurately assume the true values of the population and will also reduce the accuracy of the mixed-effects model.

Overall, these results fit well with the dual-store model of memory described in the lecture notes (Sections 1.5–1.6, pp. 8–10). According to this model, working memory is a short-term system with limited capacity that depends on rehearsal, while long-term memory stores rehearsed information for later use. Our findings show how these two systems work together and how easy it is to disturb working memory with tasks that prevent rehearsal. Even with a small group of participants, the patterns were clear and matched what cognitive theory predicts. In short, this project helps us understand how memory works in everyday life, why we remember some things easily, forget others quickly, and how attention and repetition can make the difference between short-term and long-term memory.

Sources, last visited as of 12-10-2025

1. T. S. Andersen , Lecture Notes, section 1.5
2. T. S. Andersen , Lecture Notes, section 1.6

Use of artificial intelligence

Artificial intelligence, Claude Sonnet 4.5 thinking, was used during the development of the testing program to help implement an initial draft of the experimental code. However, after the rough draft, it has undergone a huge amount of changes to help polish the code and fix any issues.

Member contribution

All group members (Laibah, Zhentao, Alex and Kaitlyn) participated actively in every part of this project, including the experimental design, data collection, code analysis, and report writing. The work was carried out collaboratively, and all members contributed equally to the final result. The main responsible students for each section is as following:

Introduction	Kaitlyn (246082)
Method	Kaitlyn (246082) and Zhentao (246213)
Results and discussion	Laibah (244320) and Kaitlyn (246082)
Conclusion	Laibah (244320)
Statistical models, python	Alex (231001)
Testing program, python	Zhentao (246213)

Code:
https://github.com/MagicBOTAlex/02464-experiments.git
Primary used files:
main.py by Zhentao for testing
dataProcess.ipynb by Zhentao for data compilation
Data-calculations.ipynb by Alex for statistical analysis and evaluation